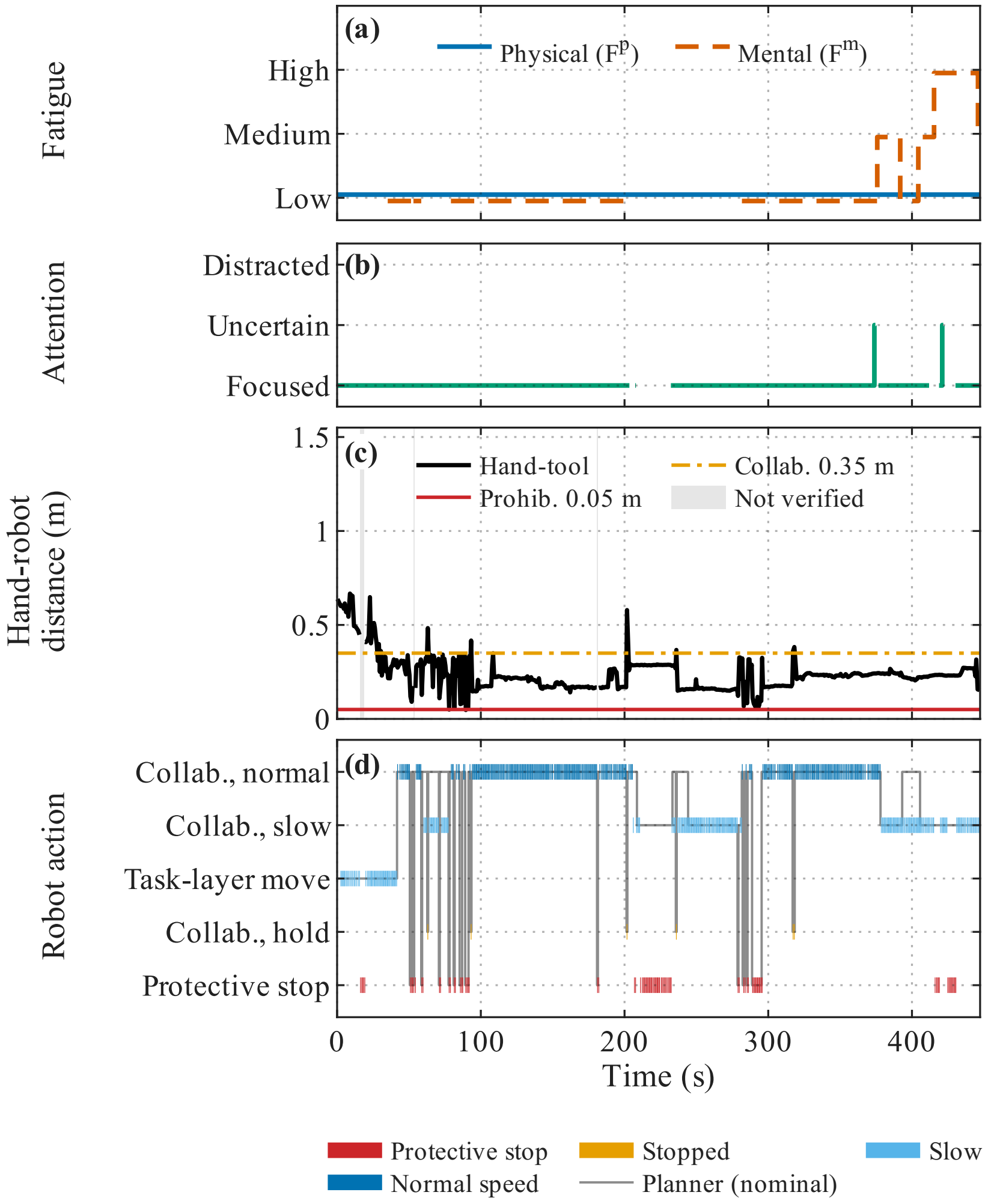

(a)
Physical (F^p)
Mental (F^m)
Fatigue
High
Medium
Low
(b)
Attention
Distracted
Uncertain
Focused
(c)
Hand-tool
Collab. 0.35 m
Prohib. 0.05 m
Not verified
Hand-robot distance (m)
1.5
1
0.5
0
(d)
Robot action
Collab., normal
Collab., slow
Task-layer move
Collab., hold
Protective stop
0
100
200
300
400
Time (s)
Protective stop
Stopped
Slow
Normal speed
Planner (nominal)